Youze (Hargen) Zheng

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LLM post-training and RL with verifiable rewards: data, compute and evaluation • AI for science

Summer 2027 research internship | available June 12 – September 20, 2027

Education

University of California San Diego — La Jolla, CA

Ph.D. in Computer Science; advisor: Ramamohan Paturi — *Sep. 2026 – Jun. 2030 (expected)*

B.S., Mathematics–Computer Science and Data Science (double major); GPA 3.94/4.00, Cum Laude — *Jun. 2026*

Publications and Preprints (* co-first authors)

[1] What Do Rollout Tokens Buy? Hidden Costs and Budgeted Accuracy in RLVR. **Y. Zheng***, et al. *Under review at ICLR 2027.*

[2] Question Begets Question: Self-Evolving Curriculum for Reinforcement Fine-Tuning on Competition Mathematics. L. Bao*, J. Wang*, Y. Zhang*, **Y. Zheng***, R. Paturi. *arXiv:2608.01522, 2026.*

[3] Single-Pass Document Scanning for Question Answering. W. Cao*, J. Wang*, **Y. Zheng***, L. Bao*, et al. *COLM 2025,* ***Oral Spotlight*** *(1.8% of submissions).* [code] [talk]

[4] CT Open: An Open-Access, Uncontaminated, Live Platform for the Open Challenge of Clinical Trial Outcome Prediction. J. Wang*, **Y. Zheng***, L. Bao*, H. Zhang*, et al. *COLM 2026.* [benchmark]

[5] DeepImagine: Clinical Trial Outcome Prediction via Stepwise Local Counterfactual Imaginations. **Y. Zheng***, J. Wang*, Y. Chen*, et al. *Under review at ICLR 2027.* [code]

[6] Measuring Risk of Bias in Biomedical Reports: The RoBBR Benchmark. J. Wang*, W. Cao*, L. Bao, **Y. Zheng**, et al. *EMNLP 2025.* [code]

Research Experience

Laboratory for Emerging Intelligence, UC San Diego — Apr. 2024 – Present

Research Assistant; advisors: Ramamohan Paturi and Leon Bergen

- **Compute-efficient RLVR [1].** Led the study; ran LoRA GRPO on Qwen3.5-9B: halving the training response limit from 4,096 to 2,048 tokens cut estimated training FLOPs **28%** and raised majority-vote accuracy on 216 held-out competition problems at 16,384 tokens per problem from **36.1% to 43.5%** as more answers finished ($+7.4 \pm 4.9$, 95% CI over problems; one run per limit). The gain held with truncated answers continued to 8,192 tokens.
- **Training data and evaluation [1].** On SmolLM3-3B (full-parameter GRPO in verl), replacing one-third of the training set with harder problems doubled response length on unchanged problems at similar training reward. At a 16,384-token budget, coverage (any correct answer) favored base SmolLM3-3B over its RL-trained version (57 vs. 39 of 216), but majority vote favored RL (18 vs. 5), as it did in all three model pairs.
- **Self-evolving RL curriculum [2].** Co-developed QbQ, which seeds new problems from those the current checkpoint mostly gets right. On the same 6,000 problems and 3,000 updates, Qwen2.5-Math-7B reached **16.46%** AIME 2025/26 pass@1 versus 14.44% with shuffled training (5.6% before training).
- **Synthetic long-document data [3].** Designed link-based synthetic QA data and trained 130M/1.3B Mamba-2 scanners on 8 H100s, diagnosing unstable gradient norms at 15k tokens; the 1.3B scanner used 15% of NV-Embed-v2-7B's retrieval FLOPs and averaged **61.8%** vs. 59.1% on 41 QA benchmarks (GPT-4o answering).
- **Clinical-trial evaluation [4].** Co-developed CT Open (live outcome-prediction benchmark) and its agentic decontamination pipeline (temporal filters, LLM web search, expert review); in a manual review of 50 random trials, 49 had no results public before the cutoff, and the one ambiguous trial was excluded.
- **Counterfactual reasoning [5].** Designed counterfactual training data from 14,000 drug RCTs, with expert-spot-checked GPT-5 reasoning traces; three Gemma-3-12B-it specialists trained with LoRA SFT reached **78.7 macro-F1** on 701 CT Open questions vs. 79.0 for GPT-5 with DeepImagine.

Projects and Skills

Post-training projects: instruction tuning and DPO on Llama-3.1-8B; chain-of-thought SFT, reward-filtered expert iteration and GRPO with verifiable rewards on MATH/GSM8K; batched vLLM evaluation harness.

ML systems (independent implementations): autodiff engine with fused operators, custom Triton GPU kernels, mixture-of-experts (tensor/expert) parallelism, MPI collectives, speculative decoding.

Languages and tools: Python, C/C++, SQL, Bash; PyTorch, verl, Hugging Face Transformers, vLLM, Triton, MPI.

Teaching and Service

Tutor/Instructional Assistant for six UC San Diego ML and data-science courses. **Reviewer,** ICLR 2027.